Energy-Efficient All-Digital Computation-in-Memory Macro Design with CFeFET-based Booth Decoder

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2026 IEEE International Symposium on Circuits and Systems (ISCAS) | 979-8-3315-7769-8/26/$31.00 ©2026 IEEE | DOI: 10.1109/ISCAS66217.2026.11562799

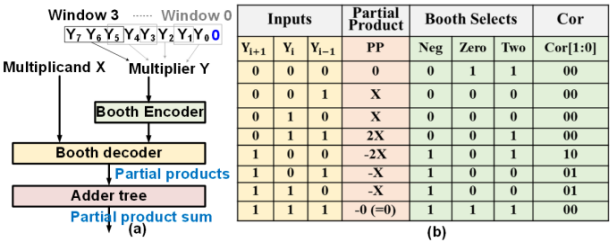
***Abstract*—The growing demand for deep neural network (DNN) accelerators has exposed the limitations of the von Neumann architecture, where the separation of memory and computation leads to excessive data movement and bandwidth constraints. Computing-in-memory (CIM) mitigates these issues by performing operations directly within memory, thereby improving energy efficiency. Compared with analog CIM, digital CIM provides superior scalability and robustness against PVT variations, yet it still suffers from large adder-tree overhead and long input interconnects. This work presents an energy-efficient all-digital CIM macro that employs the Booth algorithm to reduce partial products and alleviate adder-tree complexity. Furthermore, a complementary ferroelectric FET (CFeFET)-based Booth decoder is proposed to address the overhead introduced by Booth algorithm. To further enhance performance, a folded architecture is designed to shorten interconnect length, and voltage scaling is applied to balance speed and energy efficiency. Experimental results demonstrate that, compared with prior works, the proposed CIM macro achieves up to 5.5× higher throughput and 10.1× better energy efficiency, highlighting its potential for energy-efficient all-digital CIM accelerators.**

Keywords— Digital CIM, Booth algorithm, CFeFET -based Booth Decoder, Folded architecture

I. INTRODUCTION

DNNs have become indispensable in modern artificial intelligence (AI) applications such as image classification, segmentation, synthesis, and object detection [1-3]. The efficiency of DNNs is largely limited by intensive data movement and multiply-and-accumulate (MAC) operations, which consume a significant portion of runtime and energy [4]. However, digital accelerators suffer from excessive data transfer between processing elements (PEs) and memory, which severely limits overall energy efficiency [5-6].

Computing-in-memory (CIM) has emerged as a promising paradigm to overcome the von Neumann bottleneck by performing MAC operations directly within memory arrays [7-8]. While analog CIM offers excellent energy efficiency, its poor scalability, limited precision, and high sensitivity to PVT variations hinder practical deployment. Digital CIM (DCIM) improves robustness and CMOS compatibility but still faces challenges such as large adder-tree overhead and long input-wire loading. Prior works [9-10] have explored algorithm–circuit and algorithm–architecture co-designs to mitigate these issues. For example, [9] introduced approximation-based adders to reduce adder-tree complexity at the cost of accuracy, whereas [10] adopted Booth multiplication and reconfigurable architectures to enhance efficiency and multi-precision support. However, these approaches do not fully resolve the algorithm-induced overheads—such as the additional encoder/decoder complexity introduced by the Booth algorithm—and the long input-wire problem remains unsolved.



Inputs			Partial Product	Booth Selects			Cor
Y_{i+1}	Y_i	Y_{i-1}	PP	Neg	Zero	Two	Cor[1:0]
0	0	0	0	0	1	1	00
0	0	1	X	0	0	0	00
0	1	0	X	0	0	0	00
0	1	1	2X	0	0	1	00
1	0	0	-2X	1	0	1	10
1	0	1	-X	1	0	0	01
1	1	0	-X	1	0	0	01
1	1	1	-0 (=0)	1	1	1	00

(b)

Fig. 1. (a) Booth algorithm workflow. (b) Signals for the Booth encoder/decoder.

In this paper, we address the adder-tree and long interconnect challenges, and mitigate the overhead introduced by the Booth algorithm by proposing an energy-efficient all-digital CIM macro leveraging emerging complementary ferroelectric field-effect transistor (CFeFET) devices [11]. The proposed work achieves device–circuit–architecture–algorithm co-optimization, and the main contributions are summarized as follows:

- **Algorithm-level optimization:** The Booth algorithm is adopted for weight encoding, combined with a window-serial input scheme to reduce the number of partial products, thereby minimizing adder-tree overhead and shortening the critical path.
- **Circuit- and device-level co-optimization:** A CFeFET-based Booth decoder is proposed to address the overhead associated with Booth computation. It replaces conventional SRAM cells and CMOS decoders in DCIM macros, supporting Booth operation while achieving tighter logic–memory integration, reduced delay, and lower power consumption.
- **Architecture-level optimization:** A folded architecture with time-division multiplexing is designed to alleviate long input wires and further reducing the overhead caused by the adder tree.

II. BOOTH ALGORITHM FOR DIGITAL CIM

At the algorithm level, a Booth decoding with window-serial input scheme is employed to reduce the number of partial products, thereby minimizing the adder-tree overhead to enhance the efficiency of the proposed DCIM macro.

A. Booth algorithm with radix-4 encoding

The modified Booth algorithm [12] with radix-4 encoding is adopted to reduce partial products, where each 3-bit window produces coefficients $\{+2, +1, 0, -1, -2\}$, effectively lowering adder-tree complexity and improving computational efficiency. The relationship among encoding,

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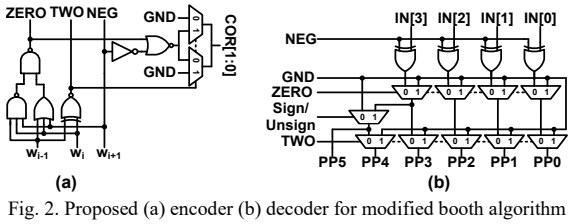


Fig. 2. Proposed (a) encoder (b) decoder for modified booth algorithm

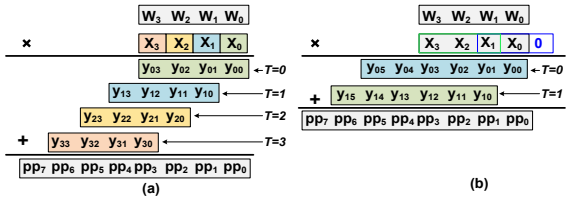


Fig. 3. (a) Bit-serial-input (b) Window-serial input for 4b MAC operation

coefficients, and generated partial products is illustrated and summarized in Fig. 1 (a) and (b).

Furthermore, we adopt a Booth encoder, shown in Fig. 2 (a), that generates four control signals—NEG, ZERO, TWO, and COR—to guide partial-product selection in the Booth decoder, presented in Fig. 2 (b). These signals correspond to the Booth encoding algorithm summarized in Fig. 1(b) , where NEG indicates negation, ZERO masks zero products, TWO represents a left-shift (×2), and COR provides the correction term for two's-complement operations.

B. Bit-Parallel MAC Operation

Fig. 3 illustrates the different serial-input for 4-bit MAC operation. Fig. 3(a) present a bit-serial-input method, which will generate 4 partial products and require 4 cycles to accumulate all partial products, with a 1-bit shift performed per cycle. By integrating the Booth algorithm, we adopt a window-serial-input 4-bit operation to achieve bit-parallel MAC computation, as shown in Fig. 3(b). In this method, a 6-bit partial product is generated per cycle, and the adder tree starts from 6 bits. It only requires two cycles to accumulate partial products, with a 2-bit shift per cycle. Furthermore, the transistor count of an eight-stage adder tree is reduced by approximately 33.25%.

C. Booth encoding for Inputs and Weights

For the DNN models, when Booth encoding is applied to the input, each 4-bit input requires four memory cells for storage. In contrast, applying Booth encoding to the weight side generates six encoded signals that must be stored. Consequently, weight-Booth encoding consumes approximately 1.5× more memory than input-Booth encoding, as presented in Fig. 4.

From the critical-path perspective, these two schemes exhibit different characteristics, as illustrated in Fig. 5. In the input-Booth approach, each input must pass through the Booth encoder during every computation cycle. Since the input value changes each cycle, the corresponding correction (COR) value also varies and must be added to the partial sum before the final accumulation. In contrast, in the weight-Booth approach, Booth encoding is pre-computed when weights are stored in memory. During computation, this removes the Booth encoder from the critical path, and the COR value remains fixed. Thus, partial sums can be

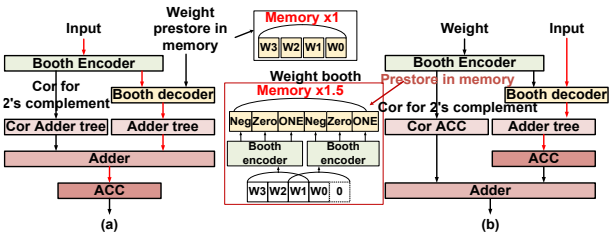


Fig. 4. Booth encoding for (a) input and (b) weights

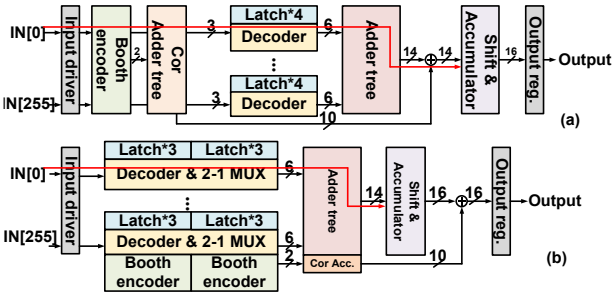


Fig. 5. Critical path for (a) IFM both (b) Weight booth

accumulated first, with the COR added only in the final stage. As a result, the weight-Booth design effectively eliminates one adder stage from the critical path, leading to improved latency and energy efficiency.

III. DCIM Macro with CFeFET-based Booth Decoder

To maintain the advantage of a shorter critical path provided by the Booth algorithm for weight encoding and to overcome its associated drawbacks, a circuit- and device-level co-optimization is proposed to further enhance the area and energy efficiency of the DCIM. In this section, a CFeFET-based Booth decoder is designed to tightly integrate computation and storage, thereby reducing area, latency, and power consumption. This co-optimization framework effectively bridges algorithmic and device innovations, achieving higher energy and area efficiency.

A. CFeFET-based MUX

As shown in Table.I and Fig. 6, the complementary FeFET (CFeFET) consists of an n-FeFET and a p-FeFET connected in series with a shared gate voltage Vin. Owing to complementary operation, only one device conducts at a time, similar to CMOS. By applying a ±3V programming pulse, the ferroelectric layer polarizes into two stable states: low-Vth /low-resistance (ON) and high-Vth/high-resistance (OFF). After programming, the polarization is retained even at 0 V, enabling non-volatile operation. The key device characteristics are summarized in Table.I.

These properties enable the device to function both as non-volatile memory and as a logic element, allowing compact implementations of logic gates and multiplexers. Leveraging these features, we implement a 2-to-1 multiplexer (2–1 MUX) using only a single n-FeFET and p-FeFET pair, as shown in Fig. 6. The source terminals are connected to Input 0 and Input 1, respectively, while the common gate serves as the selection control. During programming, a voltage pulse determines the selected input, and after programming, the device retains the state non-volatilely. This CFeFET-based MUX replaces conventional SRAM and CMOS-based MUX

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Table.I Characteristics and Operations of CFeFET

V_{write}	+3V				-3V			
	Mode	V_{th}	R	State	Mode	V_{th}	R	State
N-FeFET	Depletion	L	L	On	Enhance	H	H	Off
P-FeFET	Enhance	H	H	Off	Depletion	L	L	On

R: Resistive, L: Low, H: High

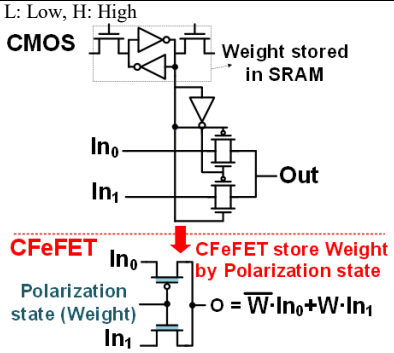


Fig. 6. Replace the SRAM and MUX by CFeFET.

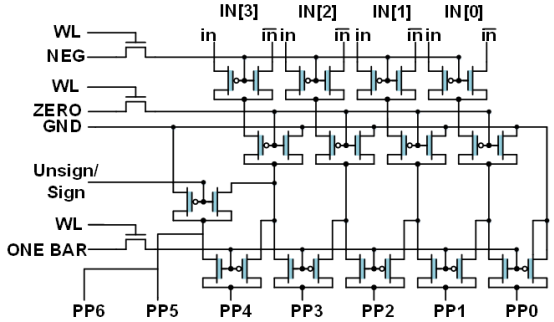


Fig. 7. Proposed CFeFET-based booth decoder

with a compact 2T implementation, achieving a significant reduction in transistor count.

B. CFeFET based Booth decoder

The CMOS-based Booth decoder, shown in Fig. 2(b), consists of MUXs and XOR gates. To enhance area efficiency, a CFeFET-based Booth decoder is proposed to replace the conventional CMOS Booth decoder and SRAM cells, as illustrated in Fig. 7. In this design, signals such as *Neg*, *Zero*, *Two*, and *Unsigned/Signed* generated from the Booth encoder are directly stored in CFeFET devices to save the SRAM cells requirement. By leveraging the compact and non-volatile properties of CFeFETs, the transistor count is reduced from 170T to 88T, resulting in a substantial area reduction and simplified routing complexity. In addition, the output signals of the CFeFET-based Booth decoder benefit from the high R_{off}/R_{on} ratio of CFeFETs, ensuring full-swing outputs with enhanced driving capability for the subsequent adder tree.

IV. Folded Structure with Time Division Multiplexing for Digital CIM

The adder tree occupies a significant portion of the area in the CIM macro. While the Booth algorithm combined with the proposed CFeFET-based Booth decoder effectively reduces the adder-tree overhead, scaling up the number of subarrays introduces longer interconnects and heavier input loading, which becomes a critical challenge for DCIM. To address this scalability issue, we propose a folded architecture with Time Division Multiplexing (TDM), which mitigates interconnect overhead and further reduces the area

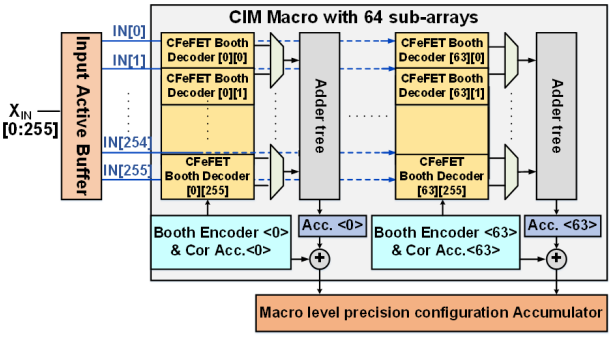


Fig. 8. The overall architecture of proposed Folded Booth Multiplication DCIM macro

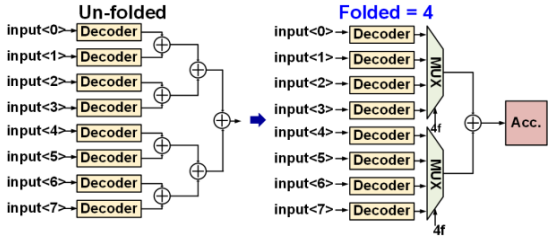


Fig. 9. The folded architecture with the time division multiplexer

and latency associated with the adder tree. This design enhances the overall scalability and energy efficiency for DCIM.

A. Overal architecture

The proposed DCIM macro consists of 64 sub-arrays, each comprising a Booth encoder, a CFeFET-based Booth decoder, MUX, an adder tree, and an accumulator, as illustrated in Fig. 8. The Booth decoder, implemented with CFeFETs, simultaneously supports weight storage and Booth multiplication, thereby enabling efficient in-memory computation. A folded architecture is employed through Time Division Multiplexing (TDM) to reduce the adder-tree area and decrease input wire length. Within each sub-array, the adder tree sums the partial products from all rows to generate partial sums, which are subsequently accumulated by the accumulator to produce the final output. The following subsections provide detailed descriptions of each component and their roles within the overall architecture.

B. Folded Architecture with Time Division Multiplexing

We design a folded architecture with TDM to reduce area and wire length by decreasing the number of adders and partitioning operations along the temporal axis through multiplexers. Although TDM increases the number of cycles required for computation, it simultaneously reduces the overall area and critical path, thereby mitigating input delay—a dominant contributor to total latency. Consequently, even with additional cycles, the total execution time can be shorter compared to designs without TDM.

In the TDM scheme, inputs are distributed through multiplexers according to selection signals. Each input is processed in different time slots, and the partial results are accumulated to reconstruct the final output from the time-distributed computations. Fig. 9 illustrates the folded structure with TDM, where four inputs share a single computation unit. While TDM introduces extra cycles, multiplexers, and accumulators, it effectively optimizes the trade-off between wire delay and cycle count, improving

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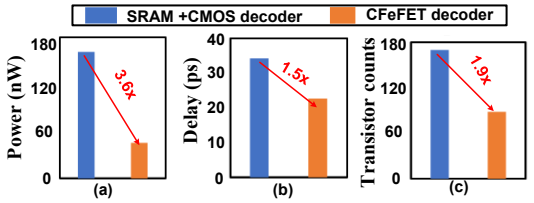


Fig. 10. Comparison between the proposed work and the conventional design for the Booth algorithm under (a) power, (b) delay, and (c) transistor count.

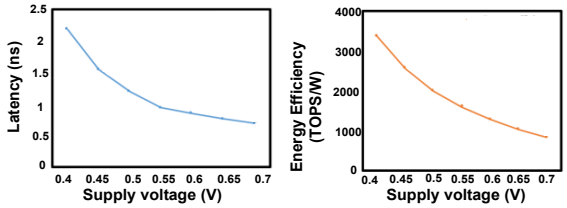


Fig. 11. The (a) latency and (b) energy-efficiency of proposed work under different voltages.

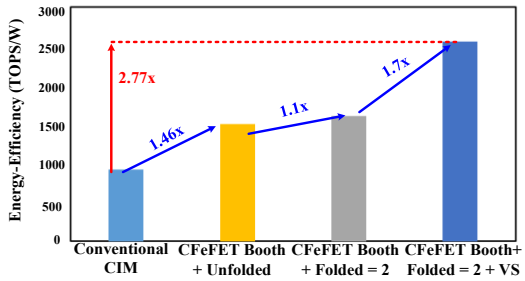


Fig. 12. The energy efficiency improvement by proposed methods

overall throughput in CIM. By mapping parallel addition operations from the spatial domain to the temporal domain, the number of adder tree stages can be reduced, expressed as:

$$\textbf{Adder tree stages}=\log_2(\text{row})-\log_2(\text{folded}) \quad (1)$$

This reduction in adder stages results in a smaller area, a shorter critical path, and improved energy efficiency in the proposed folded CIM architecture.

V. Experimental Results

The proposed work is designed and implemented using ASAP 7 nm predictive technology and CFeFET device models [11, 13]. The digital CIM macro has a capacity of 64 kb. We use SPICE simulation with the extracted parasitic components from post-layout design to evaluate energy, delay, and throughput. All simulations are conducted under nominal and scaled supply voltages to validate functionality, energy efficiency, and performance of the proposed CFeFET-based Booth decoder and folded architecture.

A. Analysis on CFeFET decoder with booth algorithm

Fig. 10 compares the power, delay, and area of the proposed design with conventional DCIM implementations using the Booth algorithm. The proposed CFeFET-based design demonstrates a notable reduction in power, delay, and area. Compared with the baseline, the proposed work achieves 3.6× lower power consumption, 1.5× shorter delay, and 1.9× reduction in transistor count. These results indicate that the proposed CFeFET-based Booth decoder not only eliminates the overhead introduced by Booth computation but also enhances energy efficiency through tighter logic–memory integration. Therefore, the combination of the Booth algorithm and CFeFET devices provides an effective cross-layer optimization path from device to algorithm.

Table. II The comparison of proposed work with other CIM accelerators

	ISSCC'20 [8]	ISSCC'21 [14]	ISSCC'22 [15]	This work
Technology	28nm	22nm	5nm	ASAP 7nm + CFeFET
CIM type	Analog	Digital	Digital	Digital
Array size	64kb	64kb	64kb	64kb
Input bit	4~8	1~8	4	4/8/16
Weight bit	4/8	4	4	4/8/16
Latency(ns)	4.1(4b/4b) 8.4(8b/8b)	10(4b/4b) 18(8b/8b)	11.1(4b/4b)	2
Throughput (GOPS)	124.8(4b/4b) 30.4(8b/8b)	3300(4b/4b) 917(8b/8b)	2950 (4b/4b)	16384(4b/4b) 4096 (8b/8b)
Energy Efficiency (TOPS/W)	68.4(4b/4b) 16.83(8b/8b)	89(4b/4b) 24.7(8b/8b)	254(4b/4b)	2553.8(4b/4b) 638.4(8b/8b)

B. Analysis on folded architecture

The latency and energy-efficiency versus voltage characteristics, presented in Fig. 11, show that the folded architecture significantly shortens the critical path, maintaining latency below 2 ns even under reduced supply voltage. This improvement enables further voltage scaling, thereby enhancing the overall energy efficiency of the proposed DCIM macro. In addition, Fig. 12 shows that the proposed design—integrating the CFeFET-based decoder, folded structure, and voltage scaling—achieves up to 2.77× higher energy efficiency than the baseline. Overall, the folded architecture provides an effective trade-off among throughput, area, and power, highlighting the potential of the proposed work for energy-efficient digital CIM.

C. Discussion

Table II compares the proposed folded CFeFET-based all-digital CIM macro with other CIM designs. Compared with recent DCIMs [8, 14, 15], the proposed work achieves 5–10× improvement in energy efficiency and 4–6× higher throughput than state-of-art digital CIMs. Compared with analog CIMs, it further provides better throughput and comparable or higher efficiency with superior precision and scalability, demonstrating strong potential for deploying DNN models.

VI. Conclusion

This work proposes a cross-layer co-optimization to improve the efficiency of digital CIM. At the algorithm level, the modified Booth algorithm with a window-serial input scheme is employed to reduce the number of partial products, thereby decreasing the area and delay overhead caused by the adder tree. Moreover, a circuit-and-device co-design approach is proposed, in which a CFeFET-based Booth decoder replaces conventional SRAM cells and CMOS decoders, mitigating the area overhead introduced by the Booth algorithm while providing lower delay and power consumption. At the architectural level, a folded architecture is designed to shorten the long interconnects of broadcast inputs and further reduce adder-tree overhead. By integrating the Booth algorithm, folded architecture, and CFeFET-based decoder, the proposed work achieves up to 5.5× higher throughput and 10.1× better energy efficiency compared with state-of-the-art designs.

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REFERENCES

[1] Z. Wang, L. Wei, T. Wang, H. Chen, Y. Hao, X. Wang, X. He and Q. Tian, “Enhance Image Classification via Inter-Class Image Mixup with Diffusion Model”, Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024.

[2] S. Minaee, Y. Boykov, F. Porikli, A. Plaza, N. Kehtarnavaz and D. Terzopoulos, "Image Segmentation Using Deep Learning: A Survey," Proc. IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 44, no. 7.

[3] X. Zhu, S. Lyu, X. Wang and Q. Zhao, “TPH-YOLOv5: Improved YOLOv5 Based on Transformer Prediction Head for Object Detection on Drone-captured Scenarios”, Proc. IEEE/CVF International Conference on Computer Vision (ICCV) Workshops, 2021.

[4] Y.-J. Chang, Y.-C. Cheng, S.-C. Liao, and C.-H. Hsiao, “A low power radix-4 booth mul tiplier with pre-encoded mechanism,” Proc. IEEE Access, vol. 8, pp. 114842–114853, 2020.

[5] Y. -H. Chen, T. -J. Yang, J. Emer and V. Sze, "Eyeriss v2: A Flexible Accelerator for Emerging Deep Neural Networks on Mobile Devices," in IEEE Journal on Emerging and Selected Topics in Circuits and Systems, vol. 9, no. 2, pp. 292-308, June 2019.

[6] K.-W. Chang and T.-S. Chang, "VWA: Hardware efficient vectorwise accelerator for convolutional neural network", IEEE Trans. Circuits Syst. I Reg. Papers, vol. 67, no. 1, pp. 145-154, Jan. 2020.

[7] Q. Dong, M. E. Sinangil, B. Erbagci, D. Sun, W.-S. Khwa, H.-J. Liao, Y. Wang, and J. Chang, “15.3 a 351tops/w and 372.4gops compute-in memory sram macro in 7nm finfet cmos for machine-learning applications,” in 2020 IEEE International Solid- State Circuits Conference - (ISSCC), 2020, pp. 242–244.

[8] [7] X. Si, Y.-N. Tu, W.-H. Huang, J.-W. Su, P.-J. Lu, J.-H. Wang, T.-W. Liu, S.-Y. Wu, R. Liu, Y.-C. Chou, Z. Zhang, S.-H. Sie, W.-C. Wei, Y.-C. Lo, T.-H. Wen, T.-H. Hsu, Y.-K. Chen, W. Shih, C.-C. Lo, R.-S. Liu, C.-C. Hsieh, K.-T. Tang, N.-C. Lien, W.-C. Shih, Y. He, Q. Li, and M.-F. Chang, “15.5 a 28nm 64kb 6t sram computing-in-memory macro with 8b mac operation for ai edge chips,” in 2020 IEEE International Solid- State Circuits Conference - (ISSCC), 2020, pp. 246–248.

[9] D. Wang et al, "DIMC: 2219TOPS/W 2569F2/b Digital In-Memory Computing Macro in 28nm Based on Approximate Arithmetic Hardware," 2022 IEEE International Solid-State Circuits Conference (ISSCC), San Francisco, CA, USA, 2022, pp. 266-268

[10] F. Tu, Y. Wang, Z. Wu, L. Liang, Y. Ding, B. Kim, L. Liu, S. Wei, Y. Xie, and S. Yin, “A28nm29.2tflops/w bf16 and 36.5tops/w int8 reconfigurable digital cim processor with unified fp/int pipeline and bitwise in-memory booth multiplication for cloud deep learn ing acceleration,” in 2022 IEEE International Solid-State Circuits Conference (ISSCC), vol. 65, 2022, pp. 1–3.

[11] Y. -Y. Huang, P. -T. Huang, P. -Y. Lee and P. Su, "A New Approach for Reconfigurable Multifunction Logic-in-Memory Using Complementary Ferroelectric-FET (CFeFET)," in IEEE Transactions on Electron Devices, vol. 70, no. 8, pp. 4497-4500, Aug. 2023.

[12] J. C. Scheunemann, M. S. Sigales, M. B. Fonseca, and E. A. C. Da Costa, “Optimizing encoder and decoder blocks for a power-efficient radix-4 modified booth multiplier,” in 2021 34th SBC/SBMicro/IEEE/ACM Symposium on Integrated Circuits and Systems De sign (SBCCI), 2021, pp. 1–6.

[13] L. T. Clark, V. Vashishtha, L. Shifren, A. Gujja, S. Sinha, B. Cline, C. Ramamurthy, and G. Yeric, “ASAP: A 7-nm finFET predictive process design kit,” Microelectronics Journal, vol. 53, pp. 105-115, Jul. 2016.

[14] Y.-D. Chih, P.-H. Lee, H. Fujiwara, Y.-C. Shih, C.-F. Lee, R. Naous, Y.-L. Chen, C.-P. Lo, C.-H. Lu, H. Mori, W.-C. Zhao, D. Sun, M. E. Sinangil, Y.-H. Chen, T.-L. Chou, K. Akarvardar, H.-J. Liao, Y. Wang, M.-F. Chang, and T.-Y. J. Chang, “16.4 an 89tops/w and 16.3tops/mm2 all-digital sram-based full-precision compute-in memory macro in 22nm for machine-learning edge applications,” in 2021 IEEE International Solid- State Circuits Conference (ISSCC), vol. 64, 2021, pp. 252–254.

[15] H. Fujiwara et al., "A 5-nm 254-TOPS/W 221-TOPS/mm2 Fully-Digital Computing-in-Memory Macro Supporting Wide-Range Dynamic-Voltage-Frequency Scaling and Simultaneous MAC and Write Operations," 2022 IEEE International Solid- State Circuits Conference (ISSCC), San Francisco, CA, USA, 2022, pp. 1-3.

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